Phú

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Tutorial 1

1.

a. S = { (head, head, head);
(head, head, tail);
(head, tail, head);
(tail, head, head);
(head, tail, tail);
(tail, head, tail);
(tail, tail, head);
(tail, tail, tail) }

b. S = { 0; 1; 2; ...; n }

c. S = { 0; 1; 2; 3; 4 }

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d.

				S
(1; Jan)	(1; Feb)	(1; Mar)	....	(1; Dec)
(2; Jan)	(2; Feb)	(2; Mar)	...	(2; Dec)
⋮	⋮	⋮		⋮
	(29; Feb)			
(31; Jan)		(31; Mar)		(31; Dec)

e.

$S = \{$(on time, satisfactorily);
(on time, unsatisfactorily);
(late, satisfactorily);
(late, unsatisfactorily)$\}$

f. $S = \{$(red; dull); (red; shine);
(blue; dull); (blue; shine)$\}$

2.

a. $\frac{p}{1-p} = 1 \Rightarrow p = \frac{1}{2}$

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b. $\frac{p}{1-p} = 2 \Rightarrow p = \frac{2}{3}$

c. $p = 0,25 \Rightarrow \frac{0,25}{1-0,25} = \frac{1}{3}$

d. $\frac{p}{1-p}$ with $0 < p < 1$

$\frac{p}{1-p} \in (0; +\infty)$

3.

a. $P(V) = 1 - P(I) - P(II) - P(III) - P(IV)$

$= 0,18$

b. $P(V) = P(IV)$

$P(V) + P(IV) = 1 - P(I) - P(II) - P(III)$

$= 0,39$

$\Rightarrow P(V) = P(IV) = 0,195$

4.

$P(I) = 2 . P(II)$

$P(II) = 3 . P(III)$

$\Rightarrow P(I) = 2 . P(II) = 6 P(III)$

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$P(I) + P(II) + P(III) = 1$

$\Leftrightarrow 6.P(III) + 3.P(III) + P(III) = 1$

$\Leftrightarrow 10.P(III) = 1$

$\Leftrightarrow P(III) = 0,1$

$\Rightarrow P(II) = 0,3$

$\Rightarrow P(I) = 0,6$

5. $P(low) + P(average) + P(high) = 1$

$\Leftrightarrow 0,28 + 0,55 + P(high) = 1$

$\Leftrightarrow P(high) = 0,17 = p$